

6502 Monitor – ESP32 Pin Mapping (NodeMCU DevKit V1)

Atari ESP32 Project

June 12, 2026

1 Data Bus (Bidirectional)

6502 Signal	ESP32 GPIO	Board Label
D0–D7	GPIO 4, 5, 13, 14, 16, 17, 18, 19	D4, D5, D13, D14, RX2, TX2, D18, D19

2 Address Bus (A0–A10, full 2 KB decode)

6502 Signal	ESP32 GPIO	Board Label
A0–A3	GPIO 34, 35, 36, 39	D34, D35, VP, VN (Input Only)
A4–A7	GPIO 32, 33, 21, 27	D32, D33, D21, D27
A8–A10	GPIO 12, 25, 26	D12, D25, D26

3 Control Signals

Signal	GPIO	Label	Dir	Description
PHI2	GPIO 2	D2	IN	6502 System Clock (1.79 MHz)
R/W	GPIO 15	D15	IN	Read/Write
SEL N	GPIO 22	D22	IN	\$D1XX / CCTL selection (Active Low)
ROMSEL	GPIO 23	D23	IN	\$D800–\$DFFF range (Active Low)
EXTSEL	GPIO 3	RX0	OUT	Disable Atari internal memory (Active Low)
MPD	GPIO 0	BOOT	OUT	Math Pack Disable (Active Low)

4 Notes

- **Full 2 KB ROM:** A0–A10 are decoded, covering all 2048 addresses of the \$D800–\$DFFF range without aliasing.
- **GPIO 0 (MPD / BOOT):** kept HIGH during boot by the Atari pull-up (isolated via TXS0108E).
- **GPIO 3 (EXTSEL / RX0):** shared with UART RX; not usable for serial input at runtime.
- All signals pass through **TXS0108E** bidirectional level shifters (3.3 V ↔ 5 V).